



Corrigendum

Corrigendum to “Numerical and experimental investigation of the effect of structural links on the sound transmission of a lightweight double panel structure”

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The authors regret two three in the paper “Numerical and experimental investigation of the effect of structural links on the sound transmission of a lightweight double panel structure” [1]. In [1], several simple models, termed “decoupled approaches”, were implemented to predict the sound transmission through a lightweight double panel partition where the panels are connected with structural junctions. One of the decoupled approaches is the model proposed by Davy [2]. Three errors were committed with that model. The first error was made in Eq. (43), which gives the velocity v_{B2} of the structural junctions on the side of the second panel

$$v_{B2} = \frac{Z_{line,1} v_1}{Z_{line,1} + [1 - (m'_B/2)C'_M \omega^2][Z_{line,2} + j\omega m'_B] + j\omega C'_M Z_{line,1} Z_{line,2} - (m'_B/2)C'_M Z_{line,1} \omega^2} \quad (43)$$

The correct expression for v_{B2} is actually

$$v_{B2} = \frac{Z_{line,1} v_1}{[Z_{line,2} + Z_{line,1}][1 - (m'_B/2)C'_M \omega^2] + j\omega C'_M Z_{line,1} Z_{line,2} + j\omega m'_B [1 - (m'_B/4)C'_M \omega^2]} \quad (43)$$

This error was made in both the manuscript and the numerical implementation. The second error was made in Eq. (48), which gives the diffuse field transmission coefficient

$$\bar{\tau}_B = \frac{64\rho_{air}^2 c_{air}^3 Q}{[g^2 + (4\omega^{1.5} m_1'' m_2'' c_{air} C'_M - g^2)^2] L \omega^2} \quad (48)$$

The correct expression for the diffuse field transmission coefficient is actually

$$\bar{\tau}_B = \frac{64\rho_{air}^2 c_{air}^3 Q}{[g^2 + (4\omega^{1.5} m_1'' m_2'' c_{air} C'_M - g)^2] L \omega^2} \quad (48)$$

This typographical error was made only in the manuscript. The third error was made in the numerical implementation of the radiation efficiency σ of the panels. In the model of Davy, this radiation efficiency is used to compute the ratio of the resonant to the non-resonant forced response of the panel. In the paper, the corrected version of Maidanik's formulae given by V  r and Holmer [3] was used to predict σ . In the multimodal region below the critical frequency (see region I.B of

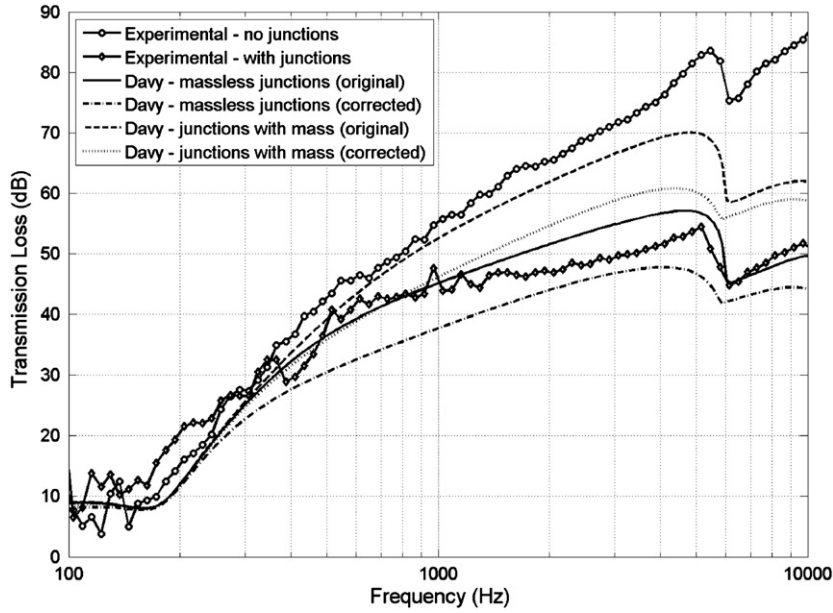


Fig. C-1. Original vs. corrected Davy curves.

table 11.3 of [3]), σ is given by

$$\sigma = \frac{2c_{\text{air}}^2}{f_c^2 S} g_1(\alpha) + \frac{P c_{\text{air}}}{f_c S} g_2(\alpha), \quad (\text{C-1})$$

where

$$\alpha = \sqrt{f/f_c}, \quad (\text{C-2})$$

$$g_1(\alpha) = \begin{cases} \frac{4}{\pi^4} \frac{(1-2\alpha^2)}{\alpha\sqrt{1-\alpha^2}} & \text{if } f < f_c/2 \\ 0 & \text{otherwise} \end{cases}, \quad (\text{C-3})$$

$$g_2(\alpha) = \frac{1}{4\pi^2} \left(\frac{(1-\alpha^2) \ln\left(\frac{1+\alpha}{1-\alpha}\right) + 2\alpha}{(1-\alpha^2)^{3/2}} \right), \quad (\text{C-4})$$

and where f is the frequency and c is the speed of sound in air. The constants f_c , S , P , L_x and L_z are the critical frequency, area, perimeter and side lengths of the panel, respectively. Instead of a factor $1/(4\pi^2)$, a factor $1/(4\pi^4)$ was wrongly used in Eq. (C-4). Therefore, the radiation efficiency below the critical frequency of the panels was underestimated, and the transmission loss (TL) of Davy's model was overestimated in Figs. 3 and 4 of the paper. The corrected TL curves have been computed and are compared in Fig. C-1 with the original curves presented in Figs. 3 and 4. The cases with and without the mass of the structural junctions are both shown in Fig. C-1. Below 1 kHz, the predicted TL when the mass of the junctions is included now agrees very well with the experimental TL. However, it is still 5–10 dB higher above that frequency. On the other hand, the predicted TL when the mass of the junctions is not included now underestimates the experimental TL by 3–10 dB over most of the frequency range. Hence, the case when the mass of the junction is included now appears more plausible. However, these results do not affect the main findings of the paper: Davy's model remains the most comprehensive model of current decoupled approaches for double panel partitions with structural junctions, but the periodic model still gives better results (see Fig. 6 of the paper).

Acknowledgments

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References

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